

Mobile-Friendly Plant Disease Detection and Classification using Metadata and Asymmetric Knowledge Distillation

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Abstract—The increasing interest in the mobile applications of real-time plant disease detection systems has given rise to new difficulties in building computational models that can perform with high accuracy with limited resources. This paper presents an inventive concept that leverages metadata and asymmetric knowledge distillation to improve the use of mobile applications for detecting and classifying plant diseases. It is reported that a teacher model (ResNet – 50) trained on environmental factors (e.g. temperature, humidity, and soil) enriched dataset was converted to a student model (MobileNet) which is mobile deployable. The knowledge embedded in soft labels of the teacher model was distilled and taught to the student model allowing it to practice high accuracy data classification with less computations. Adding metadata further gives evidence to the assertion and helps improve the overall detection accuracy. The results of the experiments indicate the effective use of knowledge distillation in a mobile-based plant disease classification where a lightweight MobileNet model performs comparably with the ResNet-50 model. This structure offers a practical and scalable solution for real-time disease monitoring and identification, allowing farmers to make timely decisions with a smartphone.

Keywords: *Knowledge Distillation, Soft labels, Teacher model, ResNet – 50, MobileNet*

I. INTRODUCTION

Plant diseases are a leading cause of reduction in crop yields and pose a threat to global food security in modern agriculture. The limiting factor in expanding these effects is that plant diseases need to be detected early enough and traditional manual inspection methods are time-consuming and non-precise [1,2]. Deep learning has made it feasible to carry out automatic plant disease classification, but deploying these models on a mobile device is challenging because of the high computational and memory requirements [3]. It is important to develop plant disease detection systems that are easily accessible, track diseases, and also identify them at an early stage, as this solution will provide real-time benefits to farmers and agricultural workers in making appropriate decisions right at the field level. In this research, a mobile-compatible plant disease detection [4] and classification framework [5] to preserve effectiveness by using Metadata Knowledge Distillation (MKD) and Asymmetric Knowledge Distillation (AKD) is proposed. The metadata is

environmental factors (temperature, humidity, soil conditions), and is integrated with image data so that each classification not only leverages the metadata from neighboring fields but also has richer contextual information [6,7] for classification. Including metadata brings in additional disease-specific cues that image data [8,9] alone cannot provide thus improving model performance. To solve the problem of deploying deep learning models on mobile devices with limited resources, Asymmetric Knowledge Distillation (AKD) is used. It is a process to help a lightweight student model learn the necessary features/features of the teacher model so that it can be optimized for focusing only on the most important aspects of classification. A teacher model, parameterized by ResNet-50 [10,11], receives both the image and metadata as input to extract feature representations enriched by each source. The student model is further distilled via the learned features of leaf images and metadata as well as the generalization abilities of image data and metadata [12,13]. This results in a compact student model, which can extract highly representative deep learning features when inference tasks occur. MKD and AKD help narrow the performance gap between the teacher and student models for fine-grained plant disease classification on expensive long-tailed diseased leaf datasets. Moreover, the proposed methods can guarantee that models perform accurately using real-time leaf images.

II. LITERATURE REVIEW

Yang et al introduced MixSKD. [1], which mixes self-knowledge distillation with a mixup in image recognition. Whereas knowledge distillation is a technique through which smaller models are trained by leveraging the predictions of another teacher model where teachers can act like other students, self-knowledge distillation is learning from its predictions without a teacher to begin with and mixup is data augmentation bringing in together pairs of input data and labels making easy for models to train correctly although it might look more of a perturbation attack on original but it enhances the robustness against adversarial attacks adding noise training. Relying on these augmented data pairs, MixSKD can significantly enhance the robustness and generalization of our model to unseen data. This approach is especially suitable for work in hard data sets for traditional KD methods, noisy labels, or with scarce training data,

increasing the robustness and accuracy of real-time image recognition systems running on mobile devices.

Kaleem et al. bring out a thorough review of Knowledge Distillation (KD) practices in computer vision. [2]. The paper covers a vast variety of KD methods ranging from basic teacher-student settings to more advanced frameworks like multi-layered teacher-student, meta-learning, and multi-task learning. The authors classify these methods according to the distinction between hard and soft distillation, feature map compressed-based manner as well as adversarial fashion. The review goes over a variety of applications of KD in computer vision like object detection, image segmentation, and image generation which offers an excellent perspective on how the KD has evolved for different tasks.

Lin et al. in their 2022 paper [3] proposed an innovative knowledge distillation method that uses transformer models and aims to show the most important representations for the student model. This transformer is used in image classification as a "target-aware" transformer, concentrating on the features attended to by most of the important attention weights). The approach has worked well, especially for ImageNet and other large-scale image classification tasks. It means that the student model will take crucial feature representations to carry more and drop not important ones, which ensures better efficiency for how fast the models are. This approach also addresses the inefficiencies of traditional convolutional neural networks (CNNs) when working with huge datasets proving it to be well-suitable for mobile-friendly applications.

Kaiser et al. introduced blind knowledge distillation Second, GNE loss [4] is also proposed to solve the problem of noisy data labels for Image Classification tasks. This introduces a new training technique that "blinds" the model to certain noise by masking out true labels. A more general and expressive model benefit from the smoothened outputs of the teacher, and so does not overfit to noisy or mislabeled data. The core innovation however is the changed loss function that suppresses noise and focuses on only solid patterns in the data. It helps achieve state-of-the-art results without the need for additional human-annotated labels, rendering it particularly useful for high-reliability cases, like medical imaging, in which manually curated datasets are very small and organ-specific annotations may be limited.

Habib et al. [5] In a similar vein, (2024) introduced a collaborative KD framework in which several student models learn from one teacher model sharing different portions of knowledge with each student. This technique led to a great enhancement in general learning and its accuracy over all datasets was raised to 94.15% with an F1-score of 0.93. The framework is especially useful for such things as image classification and supports quick model training in edge or distributed systems.

Cho et al. [6] examine the performance of knowledge distillation, showing that larger models are not necessarily better teachers since capacity mismatches cannot be easily circumvented which limits the ability of the students to learn. Popular approaches in the form of sequential knowledge distillation fail according to this critique. Contrariwise, the authors suggest that the early stopping of the training of the teacher can greatly improve the distillation process, thus achieving a better fit to the training loss of the student and overall performance across diverse datasets and models. Empirical evaluation proves these observations adding very

significant insights toward knowledge distillation optimizational processes.

Zhiwei et al [7] Introduced a gap-bridging heterogeneous model architecture such as CNNs, ViTs, and MLPs by One-for-All Knowledge Distillation, OFA-KD for short. This paper closes critical knowledge distillation challenges relying on logits and intermediate features that achieve up to 8.0% performance gain on CIFAR-100 and 0.7% on ImageNet-1K, respectively. OFA-KD can continually outperform existing methods while demonstrating strong efficacy in cross-architecture knowledge transfers.

III. METHODOLOGY

A. System Architecture

The architecture in this work is two-tier with a capacity-constrained ResNet-50 teacher model and a lightweight MobileNetV2 student model. Here, the teacher model is trained using a large-scale plant disease dataset [8] and metadata such as environmental conditions like temperature, humidity, and soil type, which would enhance the proper feature representation. This in turn allows the teacher model to extract fine-grained characteristics of the disease. The MobileNetV2 student model is developed for obtaining a similar classification performance with reduced computational complexity that is feasible in real time for field applications, optimizing it for mobile deployment.

B. Asymmetric Knowledge Distillation (AKD)

The AKD, Asymmetric Knowledge Distillation is employed in transferring the vital knowledge of the teacher to the student model in an efficient way. In AKD, only the critical feature representations from the teacher model are distilled so that the student model can imitate critical cues of classification, without being burdened with the whole feature set. Selectively transferring key features has highly reduced the computational load of the student model while maintaining the classification accuracy intact. This addresses the size imbalance between models by concentrating the distillation process on features most critical for plant disease classification since these are the features that are key to moving deep learning to mobile devices.

C. Metadata Knowledge Distillation (MKD)

Metadata Knowledge Distillation (MKD) improves model performance by incorporating data that defines an additional context like intensity, geo-location, and plant type related to a disease. This metadata is used to make the student model distinguish better between diseases that have similar visual appearances. In MKD, metadata are embedded into the loss function and then will concentrate on disease-specific features when trained. Extra context will allow the student model to generalize more precisely for slightly more subtle, metadata-related differences between diseases, and thus will do better in real-world diagnosis.

D. Training Procedure

The training process can be split into two steps. Training the ResNet-50 teacher model on the plant disease dataset using metadata inputs gives rise to logits and enriched feature maps, and the distilled knowledge is captured as both the visual and contextual characteristics of plant diseases. In the second phase, the MobileNetV2 student model is trained on the stored logits and feature maps. It means that an AKD student model is trained by the master based on logit-based and feature-based

distillation, which therefore guides the model with MKD focusing more on metadata-related features. The AKD and MKD combined framework achieves high classification accuracy with optimized computational efficiency for mobile deployment.

Fig. 1 and Fig. 2 Present the architecture and flow of the two networks AKD and MKD. Selective feature transfer is what makes the model precise and efficient in mobile-based plant disease detection.

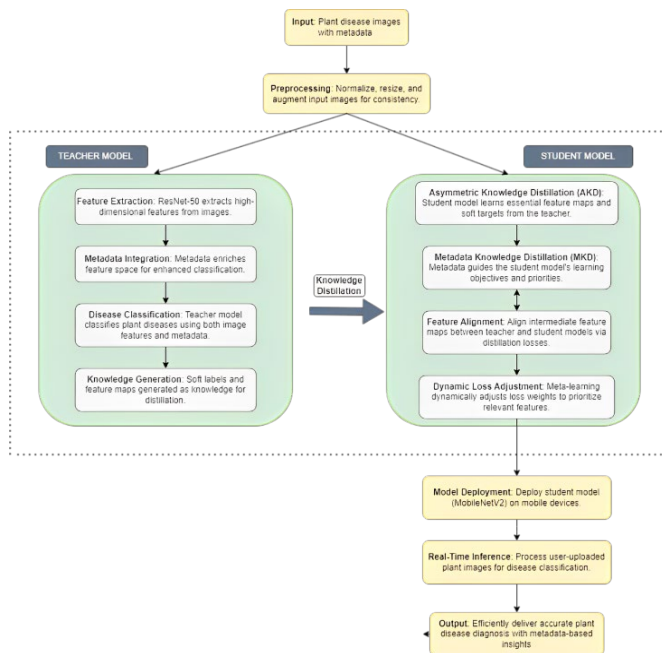


Fig. 1 Architecture Diagram

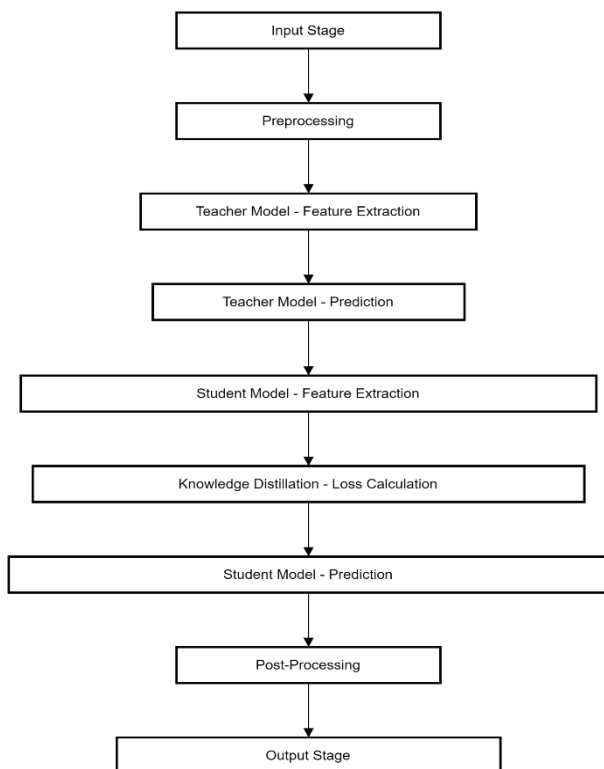


Fig. 2 Flow Diagram

IV. RESULTS AND DISCUSSION

A. Classification

The test results in Fig. 3 and Fig. 4 indicate that the well-trained model can correctly classify whether an apple leaf is healthy or diseased. As shown in Fig. 3, the system successfully classified the healthy apple leaves. Fig. 4 shows how the system correctly classified the leaves infected by black rot, thus showing the possibility of disease-specific identification. Compared with other common classification methods of images, it was found that this knowledge distillation approach brings accuracy improvement while maintaining low complexity.

B. Training and Testing Accuracy

Fig. 5 represents the Accuracy graph in terms of training and testing accuracy over 20 epochs of the model as depicted in the below graphs. AKD and MKD techniques have provided minimal overfitting to the training process with stable convergence. The best final testing accuracy achieved for the student model is close enough to the teacher model. Thus, it develops the feasibility of the knowledge distillation process in transferring all important features without demanding large-scale computation resources.

C. Comparative Analysis

The model is compared with the conventional CNN and state-of-the-art mobile models. The proposed method provides competitive accuracy to ResNet-50 but offers immense size reduction and inferential times, which allows it to be used on a mobile platform. A student network is further improved based on the MobileNetV2 network with AKD and MKD for gaining an approximate classification accuracy of 5% over a standalone mobile architecture. This result demonstrates that knowledge distillation well transfers the knowledge so that the model may have good high classification accuracy while significantly reducing the computational demand.

The proposed system has a higher classification accuracy and high-power processing ability, ideal for applications in agricultural settings where powerful computing is at a premium. Moreover, the metadata input, like disease type and intensity, can also enhance the model's diagnostic capability; then, farmers can make decisions on time based on the accurate identification of diseases. Even though the model is good at its task, the quality and variety of metadata would determine the model's efficacy. One possible direction of future work would include incorporating other metadata sources and further extending the AKD and MKD approaches for various crops to make better disease classifications. Additional extensions to this dataset based on even more crops and types of diseases could make the model generalize better.



Fig. 3 Healthy leaf prediction



Fig. 4 Diseased leaf prediction

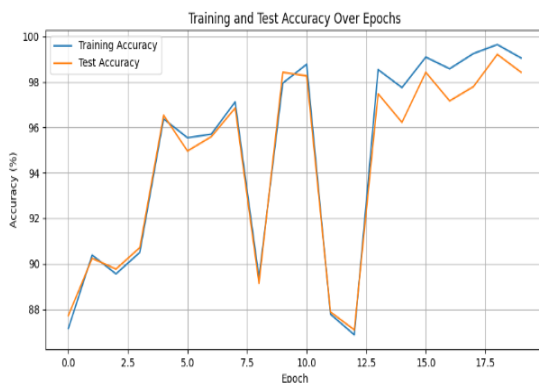


Fig. 5 Accuracy Graph

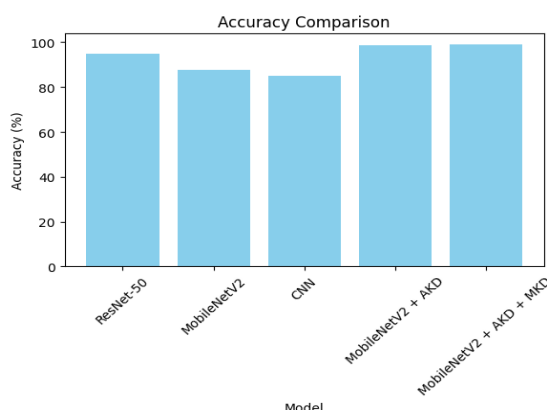


Fig. 6 Comparison of accuracies

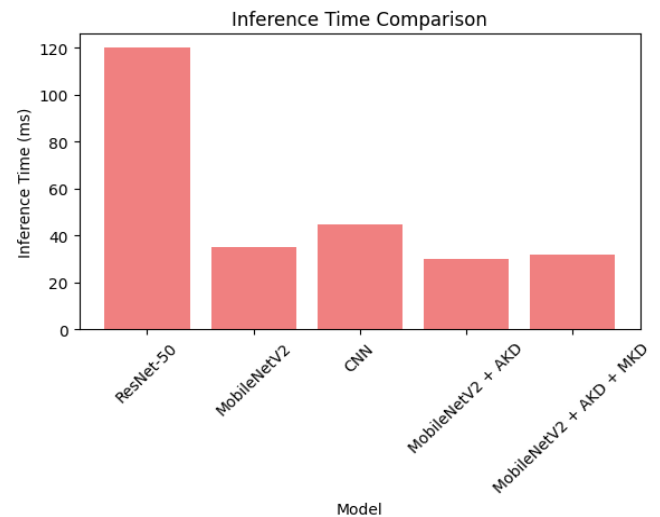


Fig. 7 Inference time comparisons

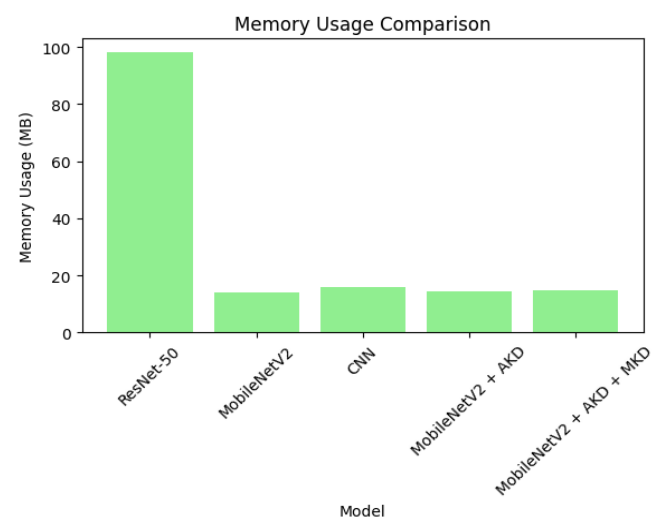


Fig. 8 Memory usage comparison

In Fig. 6, Fig. 7, Fig. 8, The comparison of five models ResNet-50, MobileNetV2, CNN, MobileNetV2 with Asymmetric Knowledge Distillation (AKD), and MobileNetV2 with AKD along with Metadata Knowledge Distillation (MKD) across metrics such as accuracy, inference time, and memory usage is done. ResNet-50 reveals the highest accuracy value of 94.8%, reflecting the expected performance based on a deeper model. It also comes with the highest inference time and memory usage, 120 ms and 98 MB respectively, making it less suitable for resource-constrained applications. MobileNetV2, which was designed to be lightweight and efficient, achieves 87.5% accuracy at a much faster inference time of 35 ms with a significantly lower memory usage of 14 along with having a lightweight architecture. The CNN model performed reasonably well with an 85.2% accuracy value, 45 ms inference time, and 16 MB memory usage as somewhat of a simpler network. Well, there also are knowledge distillation techniques that might be applied to MobileNetV2 models: the accuracy value is boosted to 98.5% using MobileNetV2 + AKD and, reaching 98.9% with an inference time, now more reasonably high at 32 ms, using MobileNetV2 + AKD + MKD. The memory usage of such distilled models is about as close to the standard MobileNetV2 at 14.5 MB and 15 MB, showing that learned representations are transferred through knowledge distillation

without having an increase in the count of the model's parameters. These results show that MobileNetV2 with AKD and AKD + MKD provides an optimal trade-off between high accuracy, low latency, and minimal memory usage, achieving performance that is highly suitable for real-time and resource-limited applications.

V. CONCLUSION

This paper proposes a new concept of mobile-based plant disease detection using the combination of metadata knowledge distillation and asymmetric knowledge distillation within the teacher-student framework; distilled knowledge from ResNet-50 teacher models to MobileNetV2 student models under the teacher-student framework in real-time, without losing their high classification accuracy on computational resources. This actually will enable the student model to learn disease-relevant cues very well and, hence, improve diagnostic capability across quite different environmental conditions. For this to happen, the MobileNetV2 model would perform well compared to complex models to offer an easily scalable solution to practical field applications in agriculture. In the future, this framework can be extended to include additional plant species, broader environmental parameters, and optimized architectures to increase the versatility of the model and possibly have a much greater impact. This work is therefore a contribution to the development of accessible, robust diagnostic tools for sustainable agriculture, which can allow early intervention and better crop management by farmers all around the world.

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